Project Plan & Study Protocol — Explainable Sepsis Risk at ED Triage

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Project summary, research questions and hypotheses

We will build an explainable model that estimates a patient's sepsis risk from information available at ED triage only, test it on public ED data, and present it as a SAFE-WAIT-style low/moderate/high traffic light for the poster competition.

Primary dataset: MIMIC-IV-ED (de-identified ED visits, Beth Israel Deaconess Medical Center, Boston), with a fully open demo subset for building the code while credentialed access is approved. **Future validation:** de-identified NSW ED data, subject to ethics and data custodian approval.

Correction to the earlier idea: the Health Gym sepsis dataset is not suitable for this question. Every patient in it already has sepsis and it is ICU data, so there are no non-sepsis patients to learn from and no triage step. It stays out of the main analysis.

RQ	Question	How we measure it	What we expect
RQ1	Can triage-only data identify patients later diagnosed with sepsis?	AUROC, AUPRC, sensitivity at a fixed alert rate	Useful discrimination from triage data alone
RQ2	Do LR, RF and XGBoost differ, and do they beat rule-based screens?	Paired bootstrap differences vs SIRS and a qSOFA-style triage rule	Tree ensembles beat LR modestly; all models beat rules at matched alert rates
RQ3	Which triage factors drive each prediction?	SHAP global and per-patient explanations	Vital-sign derangement, age, acuity, ambulance arrival and infection-type complaints dominate
RQ4	Is performance consistent across patient groups?	AUROC, sensitivity and calibration by sex, age band, race group, arrival mode	Some gaps exist; we report them rather than hide them

RQ	Question	How we measure it	What we expect
RQ5	What would a waiting-room traffic light look like in practice?	Alerts per 100 arrivals, sepsis rate per tier, net benefit	A small high-risk tier holds a large share of sepsis cases

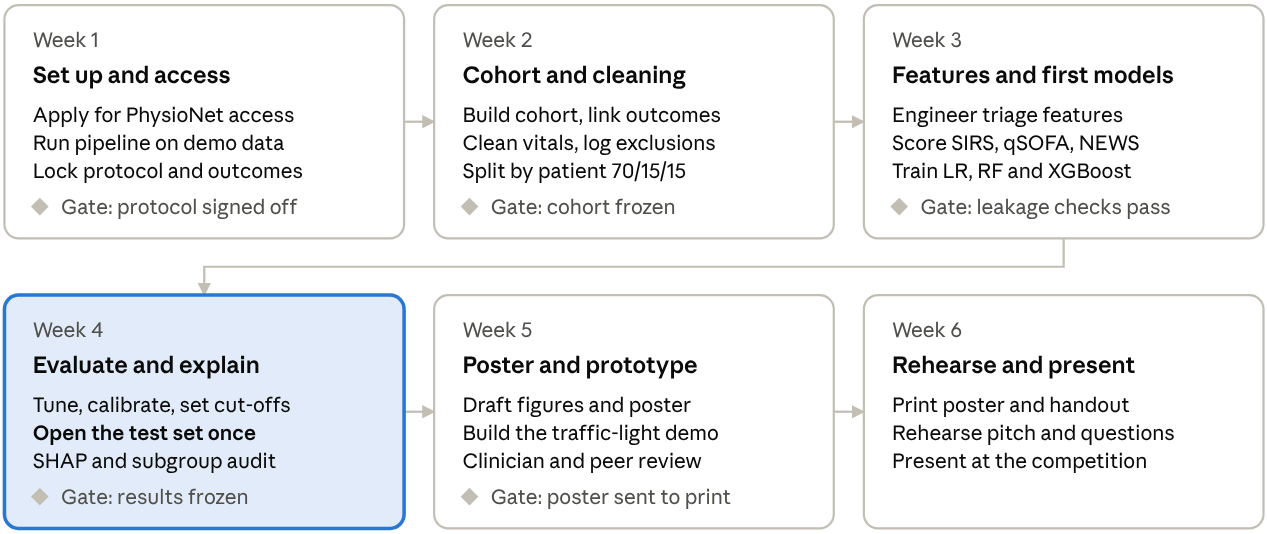
Primary outcome: a sepsis diagnosis recorded for the ED visit or its linked hospital admission. **Secondary outcome:** critical deterioration (ICU admission or in-hospital death).

Out of scope: this is a research prototype, not a medical device, and it makes no claim about NSW performance until NSW data are tested.

Timeline and phases

The work fits in six weeks if the data access request goes in on day 1; each week ends at a gate that must be passed before the next begins.

Six weeks from access request to poster, with the test set opened once in week 4



project roadmap · 6 weekly phases, 5 gates

The one step that cannot be repeated is opening the test set in week 4. Every modelling choice is fixed before it, and nothing is tuned after it.

If the deadline is closer than six weeks: merge weeks 2 and 3, and merge weeks 4 and 5. Keep every gate. If data access has not arrived by the end of week 2, switch to the protocol-poster fallback in the risks table.

Abstract timing: most competitions want the abstract before results exist. Submit the proposal-style abstract now and bring the results version to the poster.

Data sources and how to get access

The project needs one credentialed dataset for modelling, one open demo for building the code, and public NSW sources for context only. Apply for PhysioNet access on day 1, because nothing else has a waiting time.

Source	What it gives us	Role	Access
MIMIC-IV-ED v2.2	About 425,000 ED stays, 2011 to 2019: triage vitals, chief complaint, acuity, arrival mode, ED diagnoses	Primary modelling data	Credentialed: PhysioNet account, CITI "Data or Specimens Only Research" training, signed data use agreement
MIMIC-IV (hosp and icu modules)	Age, hospital diagnoses, ICU transfers, in-hospital death, past admissions	Outcomes and history features	Same credential, separate agreement to sign
MIMIC-IV-ED Demo v2.2	100 patients, same tables and columns	Build and test the code while waiting	Open, no approval
mimic4ed-benchmark	Published preprocessing code for this database	Reference to check our cleaning against	Open code
BHI Data Portal and HealthStats NSW	NSW ED volumes, triage categories, time to treatment	Context for the poster, not model training	Open
SAFE-WAIT evaluation preprint	Design and reported results of the NSW model	External reference point	Open (not yet peer reviewed)

Source	What it gives us	Role	Access
De-identified NSW ED data	Local triage records	Future validation	Ethics and custodian approval, see the governance section

Three rules that come with the data

- Every team member who opens the files needs their own PhysioNet credential. The PhysioNet FAQ says sharing within teams or classes is not permitted.
- Patient-level rows never go into ChatGPT, Claude or any online tool. PhysioNet's policy prohibits sending the data to third parties. Asking an AI tool for help with code, without data, is fine.
- Raw data stays out of GitHub. The repository holds code, the config file and aggregate results only.

Day 1 actions

- [ ] Each member: create a PhysioNet account and submit the credentialing application (it asks for a reference, usually your supervisor)
- [ ] Each member: complete the CITI course and upload the completion report
- [ ] Sign the agreements for both MIMIC-IV and MIMIC-IV-ED once credentialed
- [ ] Download both open demos and run the data stage on them. The demo is too small to train on, so run the full pipeline on the supplied stand-in data instead
- [ ] Ask your supervisor whether your institution needs an ethics exemption for public de-identified data

Cohort, outcome definitions and leakage rules

The unit of analysis is one adult ED visit, the prediction time is the end of triage, and the model may only see what a triage nurse has at that moment. Fix these three things in writing before any model is trained.

Who is included

1. Start with every row in `edstays`, joined one-to-one to `triage` on `stay_id`.
2. Keep adults: age 18 or over at the visit. Age = `anchor_age` + (visit year − `anchor_year`) from the MIMIC-IV `patients` table.

3. Drop visits with no triage vital sign recorded at all. Keep visits with partly missing vitals.
4. Record the count removed at each step. These counts become the cohort flow figure on the poster.

SAFE-WAIT excluded patients with incomplete vital signs and lists this as a selection bias. Keeping partly missing visits lets us test that group instead of hiding it.

Outcomes

Outcome	Definition	Source tables
Primary: sepsis during this episode of care	Any sepsis code in the ED diagnoses or in the linked hospital admission. ICD-10: A40, A41, R65.20, R65.21. ICD-9: 038, 995.91, 995.92, 785.52	`ed/diagnosis`, `hosp/diagnoses_icd`
Secondary: critical deterioration	In-hospital death, or ICU transfer within 12 hours of arrival. Same definition as the Xie et al. benchmark, where it occurred in 5.93% of visits	`hosp/admissions`, `icu/icustays`
Sensitivity check: Sepsis-3	Suspected infection plus a SOFA rise of 2 or more, from the MIMIC code repository's derived table. Available for ICU stays only. A stretch goal: it is not in the supplied code	`mimic-code` derived `sepsis3`

Diagnosis codes are a billing record, so they miss some true sepsis and cannot say when it began. State this on the poster. One published study on this database found 5.95% sepsis under a Sepsis-3 definition; expect a lower rate from codes alone.

What the model may and may not see

Allowed (known at triage)	Forbidden (known later, or unfair to use)
Triage temperature, heart rate, respiratory rate, oxygen saturation, blood pressure, pain score	`disposition`, `outtime`, ED length of stay

Allowed (known at triage)	Forbidden (known later, or unfair to use)
Chief complaint text	Later vital signs in `vitalsign`
Age, sex, arrival transport, hour and weekday of arrival	Medications given (`pyxis`), any laboratory result
History from before this arrival: ED visits and admissions in the past year, comorbidities coded at earlier discharges	Diagnoses from the current visit or admission
Triage acuity level, in one model variant only	Race: never a predictor, used only in the fairness audit

The code enforces the forbidden list with an automated check that fails the run if a banned column reaches the feature matrix.

How the data are split

- Split by patient, not by visit: 70% training, 15% validation, 15% test. One patient's visits never sit on both sides.
- Training is for fitting. Validation is for tuning, calibration and choosing the traffic-light cut-offs. Test is opened once, at the end.
- MIMIC shifts each patient's dates, so a true before-and-after split is not possible. Instead, the subgroup audit reports performance by `anchor_year_group` band, so drift over time is visible.

Sanity ranges from published work on the same database: triage acuity alone reached AUROC 0.80 and gradient boosting 0.88 for critical outcomes (Xie et al.); gradient boosting reached 0.83 for sepsis at triage (Liu et al. 2025). An AUROC above 0.92 on triage-only data should be treated as a leakage alarm, not a result.

Data cleaning and preparation

Cleaning turns five raw tables into one analysis table with one row per ED visit, and every rule below is written into `config.yaml` so it can be changed in one place and reported exactly.

1. **Load and link.** Read `edstays`, `triage`, `diagnosis`, `patients`, `admissions`, `diagnoses_icd` and `icustays`. Join on `stay_id`, `subject_id` and `hadm_id`.

2. **Fix temperature units.** MIMIC records triage temperature in Fahrenheit, with some values entered in Celsius by mistake. Treat 30 to 45 as Celsius, convert 86 to 113 from Fahrenheit, and set everything else to missing. NSW data will already be in Celsius.
3. **Remove impossible values.** Set a vital to missing when it falls outside the plausible range in the table below. Do not delete the visit.
4. **Clean pain and acuity.** Keep pain only when it is a number from 0 to 10. Keep acuity only when it is 1 to 5.
5. **Standardise the chief complaint.** Lower-case the text, split on commas, and map it to about 15 clinical groups with a keyword dictionary.
6. **Label outcomes.** Apply the sepsis code list and the critical deterioration rule from the previous section.
7. **Build history features.** For each visit, look back only at records with a discharge time before this arrival.
8. **Split by patient**, then fit every imputation and scaling step on the training set only.
9. **Save** the analysis table, the split assignment, and a log of row counts at each step.

Plausible ranges for triage vitals

Vital sign	Kept range	Unit
Temperature	32 to 43	°C
Heart rate	20 to 250	beats per minute
Respiratory rate	4 to 60	breaths per minute
Oxygen saturation	50 to 100	%
Systolic blood pressure	50 to 300	mmHg
Diastolic blood pressure	20 to 200	mmHg

These ranges are our starting choice. Ask a clinician to confirm them, then freeze them.

Missing data

- Report the percentage missing for every variable, overall and by outcome. Missing vitals are often informative: very sick and very well patients are both triaged quickly.
- Add a 0/1 "was missing" flag for each vital. This lets the model learn from the gap itself.
- Logistic regression and random forest get training-set medians. XGBoost receives the gaps as they are and handles them itself.

- Primary analysis keeps partly missing visits. The subgroup audit reports visits with all vitals recorded separately, as a sensitivity check.

Quality checks the code runs every time

- No patient appears in more than one split.
- No forbidden column is in the feature matrix.
- Outcome prevalence is similar across the three splits.
- There are enough cases in every split to train and test. On the 100-patient demo the run stops cleanly after the data stage.

Feature engineering

Features are built in three nested sets, so the poster can show how much each layer of triage information adds beyond vital signs alone.

Set	What it adds	Why it matters
A: Vitals	Six triage vitals, derived vital features, missing flags, age, sex	Close to the inputs the SAFE-WAIT preprint describes: vital signs, age and gender
B: Triage context	Set A plus pain, chief complaint groups, arrival mode, time of arrival, patient history	Everything else a triage nurse records or the system already knows
C: With nurse acuity	Set B plus the triage acuity level	Tests whether the model adds anything to the nurse's own judgement

Set B is the primary model. Set C is reported separately because acuity is a human judgement, and the NSW scale (the Australasian Triage Scale) differs from the US scale in MIMIC.

Feature list

Group	Features	Notes
Raw vitals	Temperature, heart rate, respiratory rate, oxygen saturation, systolic and diastolic pressure	After cleaning

Group	Features	Notes
Derived vitals	Shock index (heart rate ÷ systolic pressure), mean arterial pressure, pulse pressure	Helps logistic regression most
Clinical flags	Fever ⩾ 38.0 °C, low temperature < 36.0 °C, heart rate > 90, respiratory rate > 20 and ⩾ 22, systolic ⩽ 100, saturation < 92%, count of abnormal vitals	Mirror the cut-offs clinicians already use
Missingness	One flag per vital, plus a count	Missing is information
Demographics	Age in years, sex	Race is excluded
Arrival	Transport mode, hour of day, weekend, night	Ambulance arrival is a strong severity signal
Chief complaint	About 15 yes/no groups: fever or chills, shortness of breath, cough, abdominal pain, vomiting or diarrhoea, urinary symptoms, confusion, weakness, dizziness or collapse, chest pain, wound or skin infection, injury, and others	Keyword dictionary in the config file
History	ED visits and admissions in the previous 365 days, a previous sepsis admission, long-term conditions coded at earlier discharges (diabetes, kidney, lung, heart failure, cancer, liver, dementia)	Uses only records closed before this arrival
Acuity	Triage level 1 to 5	Set C only

Design choices to state on the poster

- Every feature is available within minutes of arrival and needs no blood test.
- The complaint dictionary is simple on purpose: a judge can read it, and it transfers to NSW free text with small edits.
- An optional extension is a bag-of-words model on the complaint text. Treat it as a stretch goal, not a core result.

Model development

Three models are trained on identical data and splits, so any difference between them comes from the algorithm and not from the setup.

Model	Role	Key settings
Logistic regression	Transparent baseline a clinician can read	Standardised inputs, L2 penalty, penalty strength tuned
Random forest	Non-linear benchmark	300 to 500 trees, minimum leaf size and feature fraction tuned
XGBoost	Main model, same family as SAFE-WAIT	Depth 3 to 6, learning rate 0.03 to 0.1, row and column subsampling, early stopping on validation

Training procedure

1. Fit all preprocessing on the training set only: medians, scaling, category encoding.
2. Tune each model with 5-fold cross-validation grouped by patient, about 25 random settings each (8 for the slower random forest), scored on area under the precision-recall curve.
3. Refit the best setting on the full training set.
4. Calibrate the predicted risks on the validation set, using isotonic regression.
5. Choose the traffic-light cut-offs on the validation set.
6. Run the locked models once on the test set and bootstrap the confidence intervals.

Handling the rare outcome

Sepsis will be a few percent of visits. We do not oversample or use SMOTE. Published work shows these corrections distort predicted risks without improving discrimination. The imbalance is handled where it belongs: in the choice of cut-off and in reporting precision-recall and alert rates.

Traffic-light cut-offs

Two cut-offs turn a risk into low, moderate or high. They are set by an alert budget a department could live with, then fixed before testing.

- High: the top 5% of arrivals by predicted risk.
- Moderate: the next 15%.

- Low: the remaining 80%.

As a second view, report the false-positive rate when sensitivity is fixed at 0.86. That is the sensitivity the SAFE-WAIT preprint reports, alongside a false-positive rate of 0.42. The populations and outcome definitions differ, so present this as context, not a head-to-head result.

Reproducibility

- One command runs the full pipeline from raw tables to figures.
- All settings live in `config.yaml`. Random seeds are fixed.
- Library versions are pinned in `requirements.txt`.
- A laptop is enough: this table is about 425,000 rows and 60 columns.

Evaluation framework and clinical baselines

A model is judged on four things: whether it ranks patients well, whether its risks are honest, whether acting on it helps, and what workload it creates. Every number is reported on the test set with a 95% confidence interval.

Question	Measure	How to read it
Does it rank sicker patients higher?	AUROC	0.5 is chance, 1.0 is perfect
Does it find the rare cases?	AUPRC	Compare with the sepsis rate, which is the chance level
How many cases does it catch at a fixed workload?	Sensitivity, specificity, PPV and NPV at the high and moderate cut-offs	The clinically meaningful numbers
What does it cost staff?	Alerts per 100 arrivals, and patients reviewed per sepsis case found (1 ÷ PPV)	The alert-fatigue numbers
Are the risks honest?	Calibration plot, slope, intercept, Brier score	A predicted 10% should mean about 10 in 100

Question	Measure	How to read it
Is acting on it worthwhile?	Decision curve: net benefit across risk thresholds of 1% to 20%	Compared with "review everyone" and "review no one"

Uncertainty and comparisons

- Confidence intervals: 1,000 bootstrap resamples of test-set patients, so repeat visits are kept together.
- Model against model, and model against rule: bootstrap the difference in AUROC and in sensitivity at a matched alert rate.
- Compare at the same alert rate. A model that flags 30% of arrivals will always look more sensitive than a rule that flags 8%.

Clinical baselines computed from the same triage data

Baseline	Triage version used here	What is missing at triage
SIRS	2 or more of: temperature > 38 or < 36 °C, heart rate > 90, respiratory rate > 20	White cell count
qSOFA	2 or more of: respiratory rate ≥ 22, systolic ≤ 100, confusion noted in the complaint	A formal mental-state score
Modified NEWS	Standard point bands for five vitals, plus confusion from the complaint	Supplemental oxygen
Triage acuity	The nurse's level on its own	Nothing: this is current practice

Label these "triage versions" on the poster. They are fair comparators for a waiting-room tool, but they are not the full bedside scores.

Reporting standard

Follow TRIPOD+AI, the 27-item reporting checklist for prediction models. A filled checklist ships with the code, and a QR code on the poster links to it.

Explainability, fairness and the traffic-light prototype

The model has to show its reasons, show who it works less well for, and show what a nurse would actually see. These three outputs are what separate this poster from a plain accuracy comparison.

Explainability

- Compute SHAP values for the XGBoost model on the test set. SHAP splits each prediction into a contribution from each feature.
- Global view: a ranked plot of the top 15 features, with direction of effect.
- Shape view: dependence plots for age, heart rate, temperature and respiratory rate. Check the curves bend where clinicians expect.
- Patient view: a "why this patient" panel listing the three strongest reasons behind one risk score.
- Cross-check: compare the SHAP ranking with logistic regression coefficients and permutation importance. Agreement across three methods is a stronger claim than SHAP alone.
- Clinical review: ask one emergency clinician or nurse whether the top drivers make sense. Quote their view on the poster.

Use an invented patient for the poster's individual example. Do not print a real MIMIC record.

Fairness audit

Subgroup	Groups
Sex	Female, male
Age	18 to 44, 45 to 64, 65 to 79, 80 and over
Race as recorded in MIMIC	Grouped into the largest categories
Arrival mode	Ambulance, walk-in
Data completeness	All vitals recorded, one or more missing

For each group report AUROC, sensitivity and false-positive rate at the high-risk cut-off, PPV, and observed versus expected sepsis rate, each with a confidence interval. Report gaps as findings. Do not tune them away.

MIMIC's race categories describe a Boston population. The NSW validation must instead examine Aboriginal and Torres Strait Islander patients, language background and remoteness, with appropriate Indigenous data governance. Say this explicitly: it is the strongest equity point available to the poster.

Traffic-light prototype

- A small local web app: enter triage vitals, age, complaint and arrival mode, and see a low, moderate or high tier with the top three reasons.
- It runs on a laptop at the poster stand with invented patients only.
- It carries a visible label: research prototype, not for clinical use.
- The summary table it rests on goes on the poster: for each tier, the share of arrivals, the sepsis rate inside it, and the share of all sepsis cases it contains.

Turning outputs into findings

Each research question maps to one output file and one sentence, so the results section is filled in rather than written from scratch. The pipeline writes every file below into the `outputs` folder.

Question	Output	Sentence to complete
Who was studied?	`cohort_flow.json`, `table1_characteristics.csv`	Of [N] adult ED visits, [n] ([x]%) had sepsis recorded.
RQ1: Does triage data work?	`table2_performance.csv`, `fig_roc_pr.png`	Using triage data alone, XGBoost reached AUROC [x] (95% CI [a to b]) and AUPRC [y], against a sepsis rate of [p]%.
RQ2: Model against rules	`table2_performance.csv`	At the same alert rate, the model found [x]% of sepsis cases and triage SIRS found [y]%.
RQ2: Model against model	`table2_performance.csv`	XGBoost exceeded logistic regression by [d] AUROC (95% CI [a to b]).

Question	Output	Sentence to complete
What does context add?	`table_feature_sets.csv`	Adding complaint, arrival and history to vitals raised AUROC from [x] to [y].
Are the risks honest?	`fig_calibration.png`	After calibration the slope was [x] and the Brier score [y].
RQ3: What drives risk?	`fig_shap_summary.png`, `table_shap_top.csv`	The strongest drivers were [1], [2] and [3].
RQ4: Is it fair?	`table4_fairness.csv`, `fig_fairness.png`	Sensitivity in the high-risk tier ranged from [x]% to [y]% across groups; the largest gap was [group].
RQ5: Would it be usable?	`table3_traffic_light.csv`, `fig_decision_curve.png`	The high-risk tier was [5]% of arrivals and held [x]% of sepsis cases, at [n] reviews per case found.

Rules for writing findings

- Lead with the clinical number: cases found per 100 alerts. Put AUROC second.
- Always give the confidence interval and the comparison at a matched alert rate.
- A small or absent gain over logistic regression is a real finding. A simple, readable model that performs nearly as well is easier to govern.
- If the model does not beat triage acuity, say so and explain what that means for NSW.
- Never write "would save lives" or "would reduce time to antibiotics". This study measures prediction, not patient outcomes.

Limitations to print on the poster

- One hospital in Boston. Performance in NSW is unknown until tested locally.
- Sepsis is defined by diagnosis codes, which under-count and carry no onset time.
- The US triage scale differs from the Australasian Triage Scale.
- Retrospective data: no effect on waiting time, antibiotics or survival is shown.
- Chief complaints are free text mapped by keywords, which will miss some wording.

Ethics, data governance and the NSW validation pathway

The current study uses public, de-identified data under a signed data use agreement; the NSW validation is a separate, later study that needs ethics and custodian approval before anyone sees a record.

This study (public data)

- Legal basis: the PhysioNet Credentialed Health Data License, signed by each team member individually.
- Institutional step: confirm with your supervisor whether an ethics exemption or notification is required for secondary use of public de-identified data.
- Storage: encrypted laptop or university secure storage. No cloud drives, no shared folders, no online AI tools for patient-level rows.
- Publication: aggregate results only. Suppress any table cell with fewer than 10 patients.

Future study (NSW data): the typical pathway

1. Find a clinical sponsor in a local health district emergency department.
2. Write a protocol and data management plan. This document is the starting draft.
3. Apply for Human Research Ethics Committee approval, usually as low-risk research with a waiver of consent for retrospective de-identified records.
4. Obtain site governance authorisation for each hospital.
5. Obtain data custodian approval. Expect the data to stay inside the health district's secure environment, with the code brought to the data.
6. Involve Aboriginal health representatives early, because the equity analysis depends on Indigenous status data.
7. If the work moves toward clinical use, complete the NSW AI Assessment Framework. High-risk systems go to the NSW AI Review Committee. Also check whether the tool counts as a medical device under TGA rules.

Confirm the exact forms with your supervisor and the health district research office. SAFE-WAIT itself ran under Western Sydney ethics approval and the NSW AI assurance approach, which makes it the model to follow.

Claims the team will and will not make

We will say	We will not say
The model estimates sepsis risk from triage data in a US dataset	The model works in NSW
It is motivated by SAFE-WAIT's clinical aim	It replicates, matches or beats SAFE-WAIT
It could support prioritisation, pending local validation	It reduces time to antibiotics or deaths
It is a research prototype	It is ready for clinical use

Team roles, risks and deliverables checklist

Five roles cover the work; a small team doubles up, and one person owns each deliverable.

Role	Owns	Main outputs
Data lead	Access, cohort, cleaning, data dictionary	Analysis table, cohort flow counts
Modelling lead	Features, training, tuning, calibration	Trained models, config file
Evaluation lead	Metrics, baselines, bootstrap, fairness audit	Performance and fairness tables, figures
Clinical and literature lead	Outcome definition, clinician review, background, references	Background text, limitations, TRIPOD+AI checklist
Poster and pitch lead	Poster, prototype demo, handout, rehearsals	Print file, pitch script, Q&A sheet

Risks and fallbacks

Risk	Likelihood	Fallback
PhysioNet credentialing is slow	Medium	Apply on day 1. Build everything on the open demo. If access misses the deadline, present a protocol poster with the validated pipeline and no performance claims
Results look too good	Medium	Run the leakage checks; compare with the published ranges for this database
Model barely beats triage acuity	Medium	Report it. Reframe around the vitals-only model and the explanation layer
No clinician available to review	Medium	Ask a nursing or medical academic at your institution; one 30-minute meeting is enough
Poster overloaded with text	High	Hold to the word limits in the poster kit
Judges ask about NSW relevance	Certain	Use the claims table and the validation pathway above

Deliverables

- [ ] Abstract submitted (plain-language and technical versions drafted)
- [ ] Data access approved for every member who handles data
- [ ] Frozen protocol: cohort, outcomes, features, cut-offs
- [ ] Analysis code running end to end from one command
- [ ] Results tables and figures from the locked test set
- [ ] Completed TRIPOD+AI checklist
- [ ] Poster print file at the size the competition specifies
- [ ] One-page handout with QR code to code and checklist
- [ ] Traffic-light prototype running offline on a laptop
- [ ] Pitch rehearsed at 30 seconds, 2 minutes and 5 minutes

Open questions for the team

- What are the abstract deadline and the competition date? The week numbers in the timeline count back from these.
- What poster size and orientation does the competition require?
- How many people are on the team, and who is the supervisor of record?

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